Dillon Corvino PhD

T cell & NK cell biology · Immuno-oncology · Single-cell & Spatial Genomics

dilloncorvino@me.com	+49 157 31659184	Bonn, Germany	github.com/Eomesodermin

PROFILE

T cell and NK cell biologist with 6+ years of postdoctoral experience across tumour immunology, viral immunity, tissue-resident immunity, and translational immuno-oncology. Combining experimental capability (primary human tissue processing, in vivo pre-clincal models, organoid systems, cellular engineering and manufacturing) with computational analysis of high-dimensional multi-omic datasets. Secured €100,000 in competitive funding as sole PI, established institue-wide spectral flow cytometry, multi-omics, and ex vivo tissue modeling. Led full scientific hiring cycles and a track record in driving projects from mecanistic discovery through peer-review publication. Now seeking to bring this depth of training to a focused industry role.

EXPERIENCE

Postdoctoral Researcher
University Hospital Bonn — Abdullah Lab
Jun 2025 – present
Institute of Molecular Medicine and Experimental Immunology — *Bonn, Germany*

- Leading independent research programme investigating unconventional T cell state transitions, tissue-resident immunity, and environmental sensing in antiviral and chronic disease contexts. Utilising LCMV infection and liver fibrosis models to identify novel immune cell programmes with translational relevance to chronic infection and inflammation
- Led full scientific recruitment cycle as primary hiring lead: authored job specifications, managed candidate pipeline, designed and executed structured interview process. Resulting in appointment of 3 PhD students and 2 research technicians; responsible for cross-functional onboarding into active experimental programmes
- Mentoring PhD and undergraduate researchers across experimental design, data analysis, and scientific writing; coordinating active international collaborations across Europe, Australia, and Japan

Scientific Consultant
Axelia Oncology — Melbourne, Australia
Jun 2023 – present

- Provided immunological data analysis, biological interpretation, and manuscript editoral support for a clinical-stage innate immunotherapy programme centred on TLR2/6 agonism (AXA-042) in solid tumours; data unpublished.

Postdoctoral Researcher
University Hospital Bonn — Bald Lab
Mar 2020 – Jun 2025
Institute of Experimental Oncology — *Bonn, Germany*

- Owned and led the NK cell cancer immunotherapy research programme. Set scientific direction, independently coordinated multi-project experimental strategy, drove execution from conception through peer-reviewed publication
- Investigated NK cell plasticity and dysfunction in the tumour microenvironment using in vivo syngeneic and humanised tumour models integrated with high-dimensional spectral cytometry. Identified cellular state transitions underlying immunotherapy resistance and defining candidate targets for combination therapy strategies
- Designed and executed all single-cell multi-omics analyses independently and end-to-end: scRNA-seq, CITE-seq, scATAC-seq, scTCR-seq, from raw sequencing data through HPC processing, QC,

integration, trajectory inference, chromatin accessibility, and biological interpretation; generated cross-functional data packages directly informing biomarker development and translational strategy across multiple collaborative programmes

- Contributed to spatial transcriptomics and high-multiplex spatial imaging analyses, including dataset design input and full downstream computational analysis, enabling tissue-resolved characterisation of immune niches in the TME
- Established primary human immune cell engineering capability in the lab. Developed and optimised CRISPR knock-out, knock-in, and viral overexpression strategies in primary human NK and CD8 T cells.
- Supervised various CAR-NK and CAR-T projects utilising established and novel CAR constructs.
- Established and operationalised three institute-wide platforms during tenture in the Bald lab: (1) spectral flow cytometry, including training all incoming staff, developing standardised panels, introducing best practices and driving institute-wide adoption. (2) single-cell multi-omics infrastructure, establishment of institute capacity to generate and analyse multi-omics datasets. (3) patient-derived tumour organoid models, implemented and co-lead the establishment of organoid program within the institute serving multiple projects within and external to the institute.
- Extensive hands-on experience with primary human tissue: processing of human tumour and healthy tissue biopsies, PBMC isolation, and generation of ex vivo model systems including air-liquid interface organoids, tissue fragment cultures, and chorioallantoic membrane (CAM) assays for testing CAR-T and CAR-NK cell therapies *in ovo*
- Secured €100,000 GO-Bio Initial BMBF grant as sole PI (2023): independently conceived, wrote, and submitted the full application for a CD56bright NK cell immunotherapy programme (ImmunoBright, code 16LW0427)
- Built and maintained cross-functional collaborations with systems biology, quantitative biology, clinical oncology, and computational groups across Germany, Australia, Japan, and the United States
- Provided day-to-day scientific leadership, experimental guidance, and manuscript support for 3 PhD candidates and multiple Masters and BSc students as co-supervisor

Postdoctoral Researcher
QIMR Berghofer Medical Research Institute — Bald Lab

Sep 2019 – Feb 2020

Oncology and Cellular Immunology Laboratory — *Brisbane, Australia*

- Single-cell profiling of tumour-infiltrating lymphocytes in head and neck squamous cell carcinoma (HNSCC); characterised inhibitory receptor landscapes and functional heterogeneity of NK and CD8 T cells, data directly informed cross-functional discussion on IO biomarker prioritisation and provided mechanistic basis for reverse translation of clinical response patterns; findings contributed to subsequent published work on NK cell dysfunction

PhD Researcher
QIMR Berghofer / University of Queensland

Apr 2015 – Sep 2019

Supervisor: Prof. Rajiv Khanna — *Thesis: Strategies for improving adoptive T-cell immunotherapy*

- Developed and characterised antigen-specific T cell therapy strategies; presented findings to technology transfer stakeholders in the context of commercialisation potential
- Contributed to clinical study linking TCR repertoire remodelling to therapeutic response in post-transplant adoptive cell therapy (JCI, 2019) — translational relevance to ACT monitoring and patient stratification

CORE COMPETENCIES

T cell & NK biology	CD8, CD4, unconventional T cell subsets, NK cells, TME biology, tissue-resident immunity, checkpoint and co-stimulatory pathways, type I IFN signalling, cytotoxicity mechanisms, resistance to cancer immunotherapy
Preclinical models	Murine syngeneic and humanised tumour models, LCMV acute and chronic viral infection, chronic liver disease, adoptive transfer, tissue immune phenotyping; chorioallantoic membrane (CAM) assay for in ovo tumour growth and immune cell response; human primary tissue processing (tumour, healthy tissue, PBMC); air-liquid interface organoids, tissue fragment cultures, and primary ex vivo culture systems

Cell engineering	CRISPR knock-out, knock-in, and overexpression in primary human NK and CD8 T cells; lentiviral and adenoviral overexpression systems; CAR-based engineering (NK and T cell); functional assay development (cytotoxicity, proliferation, ICS, degranulation)
omics	Single-cell: scRNA-seq, CITE-seq, scATAC-seq, scTCR-seq. Spatial: spatial transcriptomics and high-multiplex spatial imaging Bulk: bulk RNA-seq, bulk TCR-seq, DNA-seq, whole exome sequencing
Cytometry	Spectral and conventional multiparameter flow cytometry, FACS sorting, high-dimensional panel design (40+ colour)
Computation	R / tidyverse (primary analytical language), ggplot2, Quarto; alignment and quantification (CellRanger, STARsolo, Salmon); workflow management (Nextflow, Snakemake); reproducible compute environments (Singularity/Apptainer, Conda/Mamba); HPC with SLURM job scheduling; GitHub; single-cell QC and pre-processing (SoupX, DoubletFinder); data integration and batch correction (Harmony); imputation (MAGIC, ALRA); RNA velocity and pseudotime trajectory inference; gene regulatory network inference (SCENIC); co-expression network analysis (scWGCNA); cell-cell communication inference (CellChat, NicheNet); dataset projection and label transfer; TCR repertoire analysis (immunarch, scRepertoire); pseudobulk differential expression; mixed models and GLMMs; public dataset retrieval and deposition (GEO, ArrayExpress); large-scale public data mining (TCGA and equivalent); Python (functional proficiency); applied ML for outcome prediction and TCR structural embedding; proteomics data analysis (supporting); scientific visualisation (Adobe Illustrator)
Leadership	Independent research programme ownership; primary hiring lead (3 PhDs, 2 technicians); scientific mentorship of 10+ PhD, Masters, and BSc students; international collaboration coordination across 4 continents
Funding	PI — BMBF GO-Bio Initial €100,000 (2023); sole applicant — ImmunoSensation2 Innovation Fund €10,000 (2020)

SELECTED PUBLICATIONS

Full list: Google Scholar (899 citations, h-index 15) · ORCID: 0000-0003-0683-1369

Corvino D et al. Manuscript in preparation.
Unconventional T cell state transitions and tissue-resident immunity in antiviral infection — mechanistic characterisation with translational relevance to chronic infection and inflammatory disease

Corvino D et al. European Journal of Immunology 2025.
Type I IFN drives a dysfunctional, TCR-inert T cell state in cancer — mechanistic basis for T cell therapy failure and candidate target for improving personalised immunotherapy

Turiello R, Ng SS, … Corvino D. European Journal of Immunology 2025.
NKG7 validated as a stable TME cytotoxicity biomarker — translational utility for patient stratification and immune monitoring in clinical contexts

Braun M, … Corvino D, … Bald T. Immunity 2020.
CD155-driven degradation of CD226 as an immunotherapy resistance mechanism — informs co-stimulatory target selection and combination therapy strategy

Ng SS, … Corvino D, … Engwerda CR. Nature Immunology 2020.
NKG7 as a core regulator of cytotoxic granule exocytosis — defines an effector pathway with direct relevance to cell therapy potency and target biology

Smith C, Corvino D et al. Journal of Clinical Investigation 2019.
TCR repertoire remodelling as a correlate of clinical response to adoptive T cell therapy — supports TCR-seq-based monitoring and patient selection strategies

EDUCATION

PhD — Immunology QIMR Berghofer / University of Queensland, Australia
2015 – 2019

Bachelor of Biomedical Science, Honours Class I Griffith University, Australia
2011 – 2014
cGPA 6.83 / 7.0

FUNDING & AWARDS

GO-Bio Initial, BMBF — PI (ImmunoBright: CD56bright NK cells for cancer immunotherapy, code 16LW0427)
€100,000 · 2023

ImmunoSensation2 Postdoc Innovation Fund — Sole Applicant (Lineage tracing of NK cell plasticity)
€10,000 · 2020

Provisional Patent — Inventor ("Biomarkers of immunocompetence and uses therefor", AU 2017904567) *Filed Nov 2017*